



Progressive Education Society's
Modern College of Engineering, Pune-05.

(An Autonomous Institute Affiliated to Savitribai Phule Pune

University) Department of T.Y. B. Tech [AIDS]

Assignment No: 03

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Class: TY BTech (AIDS)
Section: A
Roll Number: 36070
Subject : Computer Vision

Title / Problem Statement:

Illustrate a tool to convert an image specifically from RGB to Grayscale and RGB to HSV color spaces, highlighting the significance of each channel.

Program Input :

```
import cv2  
  
import matplotlib.pyplot as plt  
  
def process_image(image_path):  
    # Step 3: Load the image (OpenCV loads in BGR format by default)  
    image_bgr = cv2.imread(image_path)
```

Step 4: Check if image is loaded

if image_bgr is None:

print("Error: Could not load the image. Please check the file path.")

return

Step 5: Convert BGR to RGB

image_rgb = cv2.cvtColor(image_bgr, cv2.COLOR_BGR2RGB)

Step 6: Convert RGB to Grayscale

image_gray = cv2.cvtColor(image_rgb, cv2.COLOR_RGB2GRAY)

Step 7: Convert RGB to HSV

image_hsv = cv2.cvtColor(image_rgb, cv2.COLOR_RGB2HSV)

Step 8: Split HSV channels to show their significance

h_channel, s_channel, v_channel = cv2.split(image_hsv)

Step 9: Display the results

plt.figure(figsize=(12, 8))

Original RGB

plt.subplot(2, 3, 1)

plt.imshow(image_rgb)

plt.title('Original RGB')

plt.axis('off')

Grayscale

```
plt.subplot(2, 3, 2)
plt.imshow(image_gray, cmap='gray')
plt.title('Grayscale (Intensity)')
plt.axis('off')
```

Full HSV (Visualized as RGB for structural differences)

```
plt.subplot(2, 3, 3)
plt.imshow(image_hsv)
plt.title('HSV Color Space')
plt.axis('off')
```

Hue Channel

```
plt.subplot(2, 3, 4)
plt.imshow(h_channel, cmap='gray')
plt.title('Hue Channel (Color Type)')
plt.axis('off')
```

Saturation Channel

```
plt.subplot(2, 3, 5)
plt.imshow(s_channel, cmap='gray')
plt.title('Saturation Channel (Purity)')
plt.axis('off')
```

Value Channel

```
plt.subplot(2, 3, 6)
plt.imshow(v_channel, cmap='gray')
plt.title('Value Channel (Brightness)')
```

```
plt.axis('off')
```

```
plt.tight_layout()
```

```
plt.show()
```

Run the function (Replace 'your_image.jpg' with an actual image file)

```
process_image('BMW.jpeg')
```

Program Output :

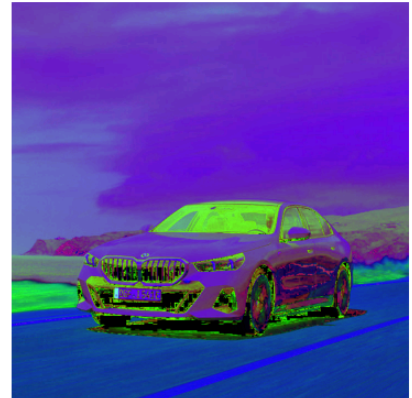
Original RGB



Grayscale (Intensity)



HSV Color Space



Hue Channel (Color Type)



Saturation Channel (Purity)



Value Channel (Brightness)

